

Homogentisate Catabolism in *Pseudomonas putida* KT2440 — Module/Pathway Review

Review topic: Homogentisate catabolism (lower/central pathway) in *Pseudomonas putida* KT2440 **Target taxon:** PSEPK — *Pseudomonas putida* KT2440 (NCBI taxon 160488; proteome UP000000556) **Queried bucket:** `kegg:ppu00643` (KEGG "Styrene degradation"), module area: aromatic_and_xenobiotic_catabolism **Candidate genes:** PP_2932, peaE/PP_3463, hmgC/PP_4619, hmgB/PP_4620, hmgA/PP_4621

1. Executive Summary

The three-reaction homogentisate central (lower) pathway is **fully present, operonic, and directly characterized** in *Pseudomonas putida* KT2440. All three expected module steps map cleanly onto the co-clustered `hmgABC` genes: **HmgA** (homogentisate 1,2-dioxygenase, EC 1.13.11.5, PP_4621/Q88E47), **HmgC** (glutathione-dependent maleylacetoacetate isomerase, EC 5.2.1.2, PP_4619/Q88E49), and **HmgB** (fumarylacetoacetate hydrolase / fumarylacetoacetase, EC 3.7.1.2, PP_4620/Q88E48). This is not an inference from homology alone: the pathway was biochemically and genetically dissected in *P. putida* by Arias-Barrau et al. (PMID: 15262943), predicted from the KT2440 genome by Jiménez et al. (PMID: 12534466), and HmgA was detected at the protein level in KT2440 by proteomics (PMID: 16470664). The module should therefore be marked **covered** with strong, species-level evidence.

The central curation finding is that the queried bucket is **the wrong container** for two of the five candidate genes. `kegg:ppu00643` resolves to the KEGG "Styrene degradation" overview map, a heterogeneous map that embeds the homogentisate branch alongside phenylacetaldehyde/phenylacetate chemistry. As a result, two genes that have nothing to do with homogentisate ring cleavage were swept into the bucket: **peaE/PP_3463** (phenylacetaldehyde dehydrogenase, feeding the phenylacetyl-CoA route) and **PP_2932** (a generic, unannotated amidase-family protein). Neither belongs to the three-reaction homogentisate module and both should be excluded from module satisfiability accounting.

Two curation caveats remain for the genuine `hmg` genes. First, the reviewed Swiss-Prot entry for HmgA (Q88E47) carries DNA-binding transcription-repressor GO terms that biologically belong to the co-encoded IclR-type regulator **HmgR**, not to the dioxygenase — a likely annotation mis-merge. Second, HmgB and HmgC are **TrEMBL (unreviewed)** entries, and HmgC lacks a recommended protein name; both warrant promotion to full gene review. The module is satisfiable today, but the bucket definition and two propagated annotations need correction.

2. Target-Organism Pathway Definition

What the pathway is

The homogentisate central pathway is a **reusable three-reaction lower pathway** that channels homogentisate (2,5-dihydroxyphenylacetate) into central carbon metabolism. In *P. putida* KT2440 the reactions are:

1. **Oxidative aromatic ring cleavage** of homogentisate → 4-maleylacetoacetate, by homogentisate 1,2-dioxygenase (HmgA, a non-heme Fe(II) enzyme).
2. **cis-trans isomerization** of 4-maleylacetoacetate → 4-fumarylacetoacetate, by maleylacetoacetate isomerase (HmgC, a glutathione-dependent GST-zeta family enzyme).
3. **Hydrolysis** of 4-fumarylacetoacetate → **fumarate** + **acetoacetate**, by fumarylacetoacetate hydrolase / fumarylacetoacetase (HmgB).

The two end products enter the TCA cycle (fumarate) and ketone-body/acetyl-CoA metabolism (acetoacetate). This is a **convergent central route**: it accepts homogentisate produced by several distinct upstream peripheral pathways.

Upstream feeders (kept separate from the module)

In *P. putida*, homogentisate is generated from **L-phenylalanine** and **L-tyrosine** (via the Phh/Hpd route: phenylalanine → tyrosine → 4-hydroxyphenylpyruvate → homogentisate, by 4-hydroxyphenylpyruvate dioxygenase, Hpd) and from **3-hydroxyphenylacetate** (via the two-component hydroxylase MhaAB in the related strain *P. putida* U) ([PMID: 15262943](#); [PMID: 15866873](#)). These upstream steps (Phh, Hpd, Mha) are **separate modules** and must not be folded into the three-reaction homogentisate lower pathway.

Boundaries and neighboring maps to keep distinct

- **KEGG "Styrene degradation" (map00643 / ppu00643)** — an overview map, *not* a clean module. It bundles the homogentisate branch with phenylacetaldehyde/phenylacetate chemistry. Membership in this map does **not** imply membership in the homogentisate module.
- **Phenylacetyl-CoA (paa) pathway** — the destination of phenylacetaldehyde dehydrogenase (peaE) product phenylacetate; a distinct catabolic route ([PMID: 20050871](#)).
- **KEGG "Tyrosine metabolism" / "Phenylalanine metabolism"** — where UniProt and KEGG place hmgABC as the terminal steps of aromatic amino-acid degradation.

Alternate names / database-specific definitions

- **HmgA**: homogentisate 1,2-dioxygenase; homogentisate oxygenase; homogentisic acid oxidase; homogentisicase; HGDO/HGD (human ortholog nomenclature). EC 1.13.11.5.
 - **HmgB**: fumarylacetoacetate hydrolase (FAH); fumarylacetoacetase. EC 3.7.1.2. (Note: gene-letter/enzyme pairing differs across databases — in KT2440 metadata, *hmgB* = the hydrolase and *hmgC* = the isomerase.)
 - **HmgC**: maleylacetoacetate isomerase (MAAI); glutathione S-transferase zeta 1 (GSTZ1) in the mammalian literature; GST-zeta family. EC 5.2.1.2.
 - In some *P. putida* genome annotations the operon is written **hmg/fah/mai** ([PMID: 12534466](#)).
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3. Expected Step Model

#	Module step	Activity (GO/EC)	Expected player	KT2440 gene	Status
1	Homogentisate oxidative ring cleavage	homogentisate 1,2-dioxygenase (EC 1.13.11.5)	HmgA	PP_4621 (hmgA, Q88E47)	Covered — direct
2	Maleylacetoacetate isomerization	maleylacetoacetate isomerase (EC 5.2.1.2)	HmgC	PP_4619 (hmgC, Q88E49)	Covered — strong
3	Fumarylacetoacetate hydrolysis	fumarylacetoacetase (EC 3.7.1.2)	HmgB	PP_4620 (hmgB, Q88E48)	Covered — strong

Step relationships (linear, product-to-substrate coupled):

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L-Phe / L-Tyr / 3-OH-phenylacetate  (upstream, SEPARATE modules)
    |
    ▼
  homogentisate
    | HmgA (PP_4621, EC 1.13.11.5, 02, Fe(II))  — STEP 1
    ▼
4-maleylacetoacetate
    | HmgC (PP_4619, EC 5.2.1.2, glutathione)  — STEP 2
    ▼
4-fumarylacetoacetate
    | HmgB (PP_4620, EC 3.7.1.2, hydrolysis)  — STEP 3
    ▼
fumarate + acetoacetate → central carbon metabolism
  
```

All three steps are encoded, contiguous, and co-regulated in KT2440 — the module is topologically complete with no orphan steps.

4. Candidate Genes and Evidence

4.1 hmgA / PP_4621 (Q88E47) — homogentisate 1,2-dioxygenase — HIGH CONFIDENCE

- **Role:** Step 1, oxidative ring cleavage of homogentisate to maleylacetoacetate. Non-heme Fe(II) dioxygenase, EC 1.13.11.5.
- **Evidence type:** *Direct, target-species.* Biochemically/genetically characterized in *P. putida* (PMID: 15262943): "*Homogentisate is then catabolized by a central catabolic pathway that involves three enzymes, homogentisate dioxygenase (HmgA), fumarylacetoacetate hydrolase (HmgB), and maleylacetoacetate isomerase (HmgC), finally yielding fumarate and acetoacetate.*" Detected at the protein level in KT2440 by 2-DE/MS proteomics, induced by phenylalanine (PMID: 16470664): "*Phenylalanine induced 4-hydroxyphenyl-pyruvate dioxygenase (Hpd) and homogentisate 1,2-dioxygenase (HmgA), key enzymes in the homogentisate degradation pathway.*" Swiss-Prot **reviewed**.
- **Curation caveat:** The Swiss-Prot entry over-merges regulator function. It carries GO terms *DNA-binding transcription repressor activity, protein-DNA complex, and negative regulation of DNA-templated transcription* alongside the correct *homogentisate 1,2-dioxygenase activity* and *iron ion binding*. The transcription-repressor terms biologically belong to the co-encoded IclR-type repressor **HmgR** (PMID: 15262943), not to the dioxygenase — a probable annotation mis-merge that should be flagged for correction.

4.2 hmgC / PP_4619 (Q88E49) — maleylacetoacetate isomerase — HIGH CONFIDENCE

- **Role:** Step 2, glutathione-dependent cis–trans isomerization of maleylacetoacetate to fumarylacetoacetate, EC 5.2.1.2. Member of the GST-zeta (GSTZ1/MAAI) family.
- **Evidence type:** *Strong; species-genomic + biochemical pathway context.* Named as one of the three central-pathway enzymes in *P. putida* (PMID: 15262943) and predicted in the KT2440 genome (PMID: 12534466). Family-level biochemistry (glutathione dependence, bifunctional GSTZ1/MAAI) is well established (PMID: 37742772).
- **Curation caveat:** TrEMBL (unreviewed); the entry has **no recommended protein name** and is annotated via GST/glutathione-transferase + maleylacetoacetate-isomerase GO. Family membership (GST superfamily) can attract broad, generic GST GO terms; ensure the specific EC 5.2.1.2 / MAAI activity is retained and generic "glutathione transferase" terms are not over-weighted. **Promote to full gene review.**

4.3 hmgB / PP_4620 (Q88E48) — fumarylacetoacetate hydrolase — HIGH CONFIDENCE

- **Role:** Step 3, hydrolysis of fumarylacetoacetate to fumarate + acetoacetate, EC 3.7.1.2 (FAH/ fumarylacetoacetase family).
- **Evidence type:** *Strong; species-genomic + biochemical pathway context.* Named as one of the three central-pathway enzymes (PMID: 15262943); part of the single `hmgABC` transcriptional unit in *P. putida*: "*the hmgABC genes appear to form a single transcriptional unit.*"
- **Curation caveat:** TrEMBL (unreviewed). FAH-family proteins are a large superfamily; confirm the specific fumarylacetoacetase activity rather than a generic FAH-domain assignment. **Promote to full gene review.**

4.4 peaE / PP_3463 (Q88H97) — phenylacetaldehyde dehydrogenase — MIS-BUCKETED (exclude)

- **Annotated role:** Phenylacetaldehyde dehydrogenase (NAD⁺), EC 1.2.1.39 — converts phenylacetaldehyde → phenylacetate, feeding the **phenylacetyl-CoA (paa)** route.
- **Why it is in the bucket:** It is a component of the KEGG "Styrene degradation" overview map (styrene → styrene oxide → phenylacetaldehyde → phenylacetate), *not* of the homogentisate branch. UniProt annotates it only with *phenylacetaldehyde dehydrogenase (NAD⁺) activity*. Homologous to the well-characterized styrene-pathway phenylacetaldehyde dehydrogenase of *P. putida* S12 (PMID: 28153386: "*Structure and biochemistry of phenylacetaldehyde dehydrogenase from the Pseudomonas putida S12 styrene catabolic pathway.*").
- **Curation action:** **Not part of the homogentisate module.** Remove from module satisfiability accounting; it belongs to a styrene/phenylacetate module.

4.5 PP_2932 (Q88IR7) — Amidase family protein — MIS-BUCKETED (exclude)

- **Annotated role:** Generic "Amidase family protein" — **no EC, no UniProt function/ pathway/GO** annotation.
- **Why it is in the bucket:** Co-occurrence in the heterogeneous KEGG map; no biochemical link to any homogentisate step.
- **Curation action:** **Not part of the homogentisate module.** Exclude. If curation attention is warranted at all, it is as an unrelated amidase, not a homogentisate enzyme.

5. Gaps, Ambiguities, and Likely Over-Annotations

No true pathway gaps. All three enzymatic steps are covered by contiguous, co-regulated genes with direct or strong species evidence. There is no missing step, no need for a lineage-specific alternative enzyme, and no step "not expected" in this organism — homogentisate catabolism is a core, experimentally confirmed capability of *P. putida* KT2440.

Bucket / boundary problem (primary issue). The `kegg:ppu00643` "Styrene degradation" map is an overview map that conflates two biochemically distinct destinations. Of the 5 candidates, **only 3 (60%)** belong to the homogentisate module; the other 2 (`peaE`, `PP_2932`) are artifacts of overview-map membership. Any module-satisfiability logic keyed on this KEGG bucket will mis-count without manual pruning.

Over-propagation / mis-merge on HmgA. The Swiss-Prot HmgA entry (Q88E47) mixes dioxygenase activity with transcription-repressor GO terms that belong to HmgR. This is a concrete example of feature mis-merge between two genes of the same operon and should be reported to UniProt curation.

Broad-family GO risk on HmgC. HmgC sits in the GST-zeta / GSTZ1 family, which in eukaryotes has pleiotropic annotations (drug metabolism, ferroptosis, cancer biomarker roles — [PMID: 37742772](#), [PMID: 36905252](#), [PMID: 35562974](#)). Those eukaryote-specific roles must **not** propagate to the bacterial enzyme; only EC 5.2.1.2 / MAAI activity is appropriate.

Regulatory conditionality (not a gap, but curation-relevant). Expression of `hmgA` / `hpd` is under **carbon catabolite (Crc) repression** in *P. putida* ([PMID: 14973036](#): "*Crc is involved in the catabolic repression of the hpd and hmgA genes from the homogentisate pathway*") and induced by homogentisate via the IclR-type repressor HmgR. The module is present but conditionally expressed; absence of expression under a given condition should not be read as absence of the genes.

6. Module and GO-Curation Recommendations

Module step	Recommended status	Rationale
1. Homogentisate ring cleavage (HmgA)	covered	Direct target-species biochemistry + KT2440 proteomics (PMID: 15262943 , PMID: 16470664)
2. Maleylacetoacetate isomerization (HmgC)	covered	Named central-pathway enzyme; genomic + family biochemistry; PP_4619 unambiguous
3. Fumarylacetoacetate hydrolysis (HmgB)	covered	Named central-pathway enzyme; operonic; PP_4620 unambiguous

Bucket-level actions:

1. **Do not treat `kegg:ppu00643` as the homogentisate module.** The clean module is the `hmgABC` operon (PP_4619–PP_4621), best associated with KEGG Tyrosine/Phenylalanine metabolism, not "Styrene degradation." Consider re-anchoring the module to the operon or to a homogentisate-specific KEGG module rather than the overview map.
2. **Remove `peaE/PP_3463` and `PP_2932` from the homogentisate module.** Re-file `peaE` under a styrene/phenylacetaldehyde→phenylacetate (`paa`) module; leave `PP_2932` as an unrelated amidase.
3. **No `not_expected_in_target_taxon` calls needed** — every module step is expected and present.
4. **No new module document strictly required** for homogentisate itself, but a separate **styrene/phenylacetaldehyde module** would cleanly absorb `peaE` and prevent future mis-bucketing from the same overview map.

GO/UniProt curation requests:

- **HmgA (Q88E47):** request removal/relocation of DNA-binding transcription-repressor GO terms (belong to HmgR); retain *homogentisate 1,2-dioxygenase activity* (GO:0004411) and *iron ion binding*.
- **HmgC (Q88E49):** add recommended protein name "Maleylacetoacetate isomerase"; ensure EC 5.2.1.2 / MAAI GO is primary; guard against generic GST or eukaryote-specific GSTZ1 terms.

- **HmgB (Q88E48)**: confirm specific fumarylacetoacetase (EC 3.7.1.2) activity over generic FAH-domain annotation.

7. Genes to Promote to Full fetch-gene Review

Gene	UniProt	Reason to promote
hmgB / PP_4620	Q88E48 (TrEMBL)	Unreviewed; verify specific fumarylacetoacetase activity vs. broad FAH family; core module step
hmgC / PP_4619	Q88E49 (TrEMBL)	Unreviewed; no recommended protein name ; GST-zeta family broad-annotation risk; core module step
hmgA / PP_4621	Q88E47 (Swiss-Prot)	Reviewed, but flag the HmgR transcription-repressor GO mis-merge for correction

peaE/PP_3463 and PP_2932 do **not** warrant homogentisate-module review; they should be re-routed out of this bucket.

8. Key References

- Arias-Barrau E, et al. *The homogentisate pathway: a central catabolic pathway involved in the degradation of L-phenylalanine, L-tyrosine, and 3-hydroxyphenylacetate in Pseudomonas putida*. J Bacteriol. 2004. PMID: [15262943](#) — **Primary direct evidence**: defines HmgA/HmgB/HmgC, the operon, and HmgR regulation.
- Jiménez JI, et al. *Genomic analysis of the aromatic catabolic pathways from Pseudomonas putida KT2440*. Environ Microbiol. 2002. PMID: [12534466](#) — Genome-level prediction of the homogentisate (hmg/fah/mai) pathway in the exact target strain.
- Kim YH, et al. *Analysis of aromatic catabolic pathways in Pseudomonas putida KT2440 using a combined proteomic approach: 2-DE/MS and cleavable isotope-coded affinity tag analysis*. Proteomics. 2006. PMID: [16470664](#) — Protein-level detection of HmgA induction in KT2440.
- Morales G, et al. *The Pseudomonas putida Crc global regulator controls the expression of genes from several chromosomal catabolic pathways for aromatic compounds*. J Bacteriol. 2004. PMID: [14973036](#) — Crc catabolite repression of hmgA/hpd.

- Herrera MC & Ramos JL. *Identification and characterization of the PhhR regulon in Pseudomonas putida*. 2010. [PMID: 20050871](#) — Upstream Phe/Tyr regulon and the distinct paa route (context for peaE mis-bucketing).
 - Arias-Barrau E, et al. *A two-component hydroxylase involved in the assimilation of 3-hydroxyphenyl acetate in Pseudomonas putida*. 2005. [PMID: 15866873](#) — Upstream 3-OH-phenylacetate → homogentisate feeder (MhaAB); HmgABC as convergent central route.
 - Crabo AG, et al. *Structure and biochemistry of phenylacetaldehyde dehydrogenase from the Pseudomonas putida S12 styrene catabolic pathway*. 2017. [PMID: 28153386](#) — Confirms peaE-type enzyme belongs to the styrene/phenylacetate route, not homogentisate.
 - Board PG. *Clinical physiology and pharmacology of GSTZ1/MAAI*. 2023. [PMID: 37742772](#) — MAAI/GSTZ1 family biochemistry relevant to HmgC annotation.
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Bottom line for curation

Mark all three homogentisate module steps **covered** in *P. putida* KT2440 on the strength of the operonic `hmgABC` genes (PP_4619–PP_4621) with direct species evidence. Prune peaE/PP_3463 and PP_2932 from the bucket as overview-map artifacts. Promote HmgB and HmgC (TrEMBL) to full gene review, and flag the HmgA→HmgR transcription-repressor GO mis-merge for correction.
